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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/134,756	12/28/2020	SIDDHARTH RAVICHANDRAN	200397-US-ORG1	3046

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EXAMINER	
BOYLE, ABBIGALE A	

ART UNIT	PAPER NUMBER
2899	

NOTIFICATION DATE	DELIVERY MODE
04/10/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/134,756	Applicant(s) RAVICHANDRAN et al.	
	Examiner ABBIGALE A BOYLE	Art Unit 2899	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12 February 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 9-10,12,23-29 and 36-45 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 9-10,12,23-29 and 36-45 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 28 December 2020 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 12 February 2025 has been entered.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 9-11, 23, 25, 28-29, 36-37, 39, 41, and 44-45 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860).

Regarding Claim 9, Tai et al. discloses an apparatus with optical fiber attach capability, the apparatus comprising:

a substrate (substrate 122, Figure 44);

a system on a chip disposed on the substrate (chip 50a, Figure 44);

a photonic integrated circuit with a v-groove in a front side fiber coupling region (PIC 50c, groove 76, Figure 44);

an organic redistribution layer positioned between the system on a chip and the substrate and communicating with the chip and photonic integrated circuit (organic redistribution layer 68/66 or 68/64/62 or 68/66/62 or 68/64/62/66 or 60/62/64/68 or 60/62/64/68/66, Figures 1, 39, and 44. In other words, the RDL is 68 in combination with at least one metal layer and optionally with other adjacent metal and insulating layers); and

an optical fiber attached to the front side fiber coupling region (optical fiber 168, Figure 44),

wherein a mold compound encapsulates the system on a chip and the photonic integrated circuit (mold compound 120, Figure 44), and

wherein the substrate includes a cutout at the attachment of the optical fiber to the photonic integrated circuit (substrate 122, Figure 44).

However, they do not explicitly disclose that the mold compound includes a cutout at the attachment of the optical fiber to the photonic integrated circuit. Chen et al. discloses a similar device wherein a mold compound encapsulates a chip, a photonic integrated circuit with a groove in a front side, said the mold compound including a cutout at the attachment of the optical fiber to the photonic integrated circuit (chip 120, photonic

integrated circuit 110c, optical fiber 160c, mold compound 130, Figure 15). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to include a cutout in the mold compound at the attachment of the optical fiber to the photonic integrated circuit in Tai et al. in view of Chen et al. in order to simplify the complexity of manufacturing and reduce production cost (Chen et al., Paragraph 38).

Regarding Claim 10, Tai et al. in view of Chen et al. further disclose wherein the apparatus is a die-last wafer- level fanout package (Tai et al., package 601, Figure 44). The language, term, or phrase "die-last wafer-level fanout package," is directed towards the process of making a wafer-level fanout package. It is well settled that "product by process" limitations in claims drawn to structure are directed to the product, per se, no matter how actually made. In re Hirao, 190 USPQ 15 at 17 (footnote 3). See also, In re Brown, 173 USPQ 685; In re Luck, 177 USPQ 523; In re Fessmann, 180 USPQ 324; In re Avery, 186 USPQ 161; In re Wethheim, 191 USPQ 90 (209 USPQ 554 does not deal with this issue); In re Marosi et al., 218 USPQ 289; and particularly In re Thorpe, 227 USPQ 964, all of which make it clear that it is the patentability of the final product per se which must be determined in a "product by process" claim, and not the patentability of the process, and that an old or obvious product produced by a new method is not patentable as a product, whether claimed in "product by process" claims or otherwise. The above case law further makes clear that applicant has the burden of showing that the method language necessarily produces a structural difference. As such, the language "die-last wafer-level fanout package" only requires a wafer-level fanout package with a die, which

does not distinguish the invention from Tai et al. in view of Chen et al., who teaches the structure as claimed.

Regarding Claim 23, Tai et al. in view of Chen et al. further disclose wherein the mold compound comprises an epoxy material (Tai et al., mold compound 120, Figure 44, Paragraph 36).

Regarding Claim 25, Tai et al. in view of Chen et al. further disclose wherein the organic redistribution layer comprises at least one polymer layer (Tai et al., Paragraph 24) and that the organic redistribution layer comprises a plurality of insulating layers (Tai et al., Paragraphs 20 and 24) but they do not explicitly disclose that the organic redistribution layer comprises a plurality of polymer layers. Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the organic redistribution layer to comprise a plurality of polymer layers in Tai et al. in view of Chen et al. in order to use suitable known dielectric/passivation materials.

Regarding Claim 28, Tai et al. in view of Chen et al. further disclose wherein the organic redistribution layer is attached to the substrate with a plurality of bumps (Tai et al., RDL 68/64/62/60, bumps 66, Figures 1 and 44, paragraph 22-23).

Regarding Claim 29, Tai et al. in view of Chen et al. does not explicitly disclose wherein the plurality of bumps comprises a ball grid array. However, they disclose the use of a ball grid array (Tai et al., BGA 150, Figure 44, Paragraph 051) and the need for

increasing component density and level of integration (Tai et al., Paragraph 002). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the bumps as a grid and therefore have the plurality of bumps comprise a ball grid array in Tai et al. in view of Chen et al. in order to increase component density.

Regarding Claim 36, Tai et al. discloses a system with optical fiber attach capability, the system comprising:

a semiconductor chip package including:

a substrate (substrate 122, Figure 44)

a system on a chip disposed on the substrate (chip 50a, Figure 44);

a photonic integrated circuit with a v-groove in a front side fiber coupling region (PIC, 50c, groove 76, Figure 44);

an organic redistribution layer positioned between the system on a chip and the substrate and communicating with the chip and photonic integrated circuit organic redistribution layer 68/66 or 68/64/62 or 68/66/62 or 68/64/62/66 or 60/62/64/68 or 60/62/64/68/66, Figures 1, 39, and 44. In other words, the RDL is 68 in combination with at least one metal layer and optionally with other adjacent metal and insulating layers); and

an optical fiber attached to the front side fiber coupling region (optical fiber 168, Figure 44),

wherein a mold compound encapsulates the system on a chip and the photonic integrated circuit (mold compound 120, Figure 44),

wherein the substrate include a cutout at the attachment of the optical fiber to the photonic integrated circuit (substrate 122, Figure 44).

However, they do not explicitly disclose that the mold compound includes a cutout at the attachment of the optical fiber to the photonic integrated circuit.

Chen et al. discloses a similar device wherein a mold compound encapsulates a chip, a photonic integrated circuit with a groove in a front side, and an optical fiber, said the mold compound including a cutout at the attachment of the optical fiber to the photonic integrated circuit (chip 120, photonic integrated circuit 110c, optical fiber 160c, mold compound 130, Figure 15). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to include a cutout in the mold compound at the attachment of the optical fiber to the photonic integrated circuit in Tai et al. in view of Chen et al. in order to simplify the complexity of manufacturing and reduce production cost (Chen et al., Paragraph 38).

Regarding Claim 37, Tai et al. in view of Chen et al. further disclose wherein the apparatus is a die-last wafer- level fanout package (Tai et al., package 601, Figure 44). The language, term, or phrase "die-last wafer-level fanout package," is directed towards the process of making a wafer-level fanout package. It is well settled that "product by process" limitations in claims drawn to structure are directed to the product, per se, no matter how actually made. In re Hirao, 190 USPQ 15 at 17 (footnote 3). See also, In re Brown, 173 USPQ 685; In re Luck, 177 USPQ 523; In re Fessmann, 180 USPQ 324; In re Avery, 186 USPQ 161; In re Wethheim, 191 USPQ 90 (209 USPQ 554 does not deal with this issue); In re Marosi et al., 218 USPQ 289; and particularly In re Thorpe, 227

USPQ 964, all of which make it clear that it is the patentability of the final product per se which must be determined in a "product by process" claim, and not the patentability of the process, and that an old or obvious product produced by a new method is not patentable as a product, whether claimed in "product by process" claims or otherwise. The above case law further makes clear that applicant has the burden of showing that the method language necessarily produces a structural difference. As such, the language "die-last wafer-level fanout package" only requires a wafer-level fanout package with a die, which does not distinguish the invention from Tai et al. in view of Chen et al., who teaches the structure as claimed.

Regarding Claim 39, Tai et al. in view of Chen et al. further disclose wherein the mold compound comprises an epoxy material (Tai et al., mold compound 120, Figure 44, Paragraph 36).

Regarding Claim 41, Tai et al. in view of Chen et al. further disclose wherein the organic redistribution layer comprises at least one polymer layer (Tai et al., Paragraph 24) and that the organic redistribution layer comprises a plurality of insulating layers (Tai et al., Paragraphs 20 and 24) but they do not explicitly disclose that the organic redistribution layer comprises a plurality of polymer layers. Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the organic redistribution layer to comprise a plurality of polymer layers in Tai et al. in view of Chen et al. in order to use suitable known dielectric/passivation materials.

Regarding Claim 44, Tai et al. in view of Chen et al. further disclose wherein the organic redistribution layer is attached to the substrate with a plurality of bumps (Tai et al., RDL 68/64/62/60, bumps 66, Figures 1 and 44, paragraph 22-23).

Regarding Claim 45, Tai et al. in view of Chen et al. does not explicitly disclose wherein the plurality of bumps comprises a ball grid array. However, they disclose the use of a ball grid array (Tai et al., BGA 150, Figure 44, Paragraph 051) and the need for increasing component density and level of integration (Tai et al., Paragraph 002). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the bumps as a grid and therefore have the plurality of bumps comprise a ball grid array in Tai et al. in view of Chen et al. in order to increase component density.

Claims 12 and 24 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 9 above, further in view of Weng et al. (U.S. 2022/0365297).

Regarding Claim 12, Tai et al. in view of Chen et al. does not explicitly disclose that the attached fiber is secured by a glob top. However, they disclose the optical fiber is coupled to the package via an adhesive layer (Tai et al., fiber 168, adhesive layer 166, Figure 5, Paragraph 61). Weng et al. discloses a similar device wherein an optical fiber is secured to a PIC that is part of an encapsulated package by a glob top (PIC 50B, optical fiber 168a, glob top 166, Figure 13). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to secure the attached fiber by a glob

top in Tai et al. in view of Chen et al., further in view of Weng et al. in order enhance coupling efficiency (Weng et al., Paragraph 63).

Regarding Claim 24, Tai et al. in view of Chen et al, further in view of Weng et al. do not explicitly disclose wherein the glob top comprises an epoxy material. Tai et al. disclose that the molding material is an epoxy material (Tai et al., Paragraph 36), Chen et al. discloses attaching an optical fiber using an epoxy material (chip 120, PIC 110, grooves 112, fiber 160, RDL 140, Figures 14 and 15, Paragraph 17), and Weng et al. discloses that the glob top of a light-transmissive curable gel (Weng et al., Paragraph 63). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the glob top to comprise an epoxy material in Tai et al in view of Chen et al., further in view of Weng et al., in order to facilitate an optical interface using a suitable optical adhesive to provide optical transparency and mechanical fixation (Chen et al., Paragraph 17). Furthermore it has been held that the selection of a known material based on its suitability for its intended use supported a prima facie obviousness determination in Sinclair & Carroll Co. v. Interchemical Corp., 325 U.S. 327, 65 USPQ 297 (1945), see MPEP 2144.07.

Claims 38 and 40 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 36 above, further in view of Weng et al. (U.S. 2022/0365297).

Regarding Claim 38, Tai et al. in view of Chen et al. does not explicitly disclose that the attached fiber is secured by a glob top. However, they disclose the optical fiber

is coupled to the package via an adhesive layer (Tai et al., fiber 168, adhesive layer 166, Figure 5, Paragraph 61). Weng et al. discloses a similar device wherein an optical fiber is secured to a PIC that is part of an encapsulated package by a glob top (PIC 50B, optical fiber 168a, glob top 166, Figure 13). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to secure the attached fiber by a glob top in Tai et al. in view of Chen et al., further in view of Weng et al. in order enhance coupling efficiency (Weng et al., Paragraph 63).

Regarding Claim 40, Tai et al. in view of Chen et al, further in view of Weng et al. do not explicitly disclose wherein the glob top comprises an epoxy material. Tai et al. disclose that the molding material is an epoxy material (Tai et al., Paragraph 36), Chen et al. discloses attaching an optical fiber using an epoxy material (chip 120, PIC 110, grooves 112, fiber 160, RDL 140, Figures 14 and 15, Paragraph 17), and Weng et al. discloses that the glob top of a light-transmissive curable gel (Weng et al., Paragraph 63). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to form the glob top to comprise an epoxy material in Tai et al in view of Chen et al., further in view of Weng et al., in order to facilitate an optical interface using a suitable optical adhesive to provide optical transparency and mechanical fixation (Chen et al., Paragraph 17). Furthermore it has been held that the selection of a known material based on its suitability for its intended use supported a prima facie obviousness determination in Sinclair & Carroll Co. v. Interchemical Corp., 325 U.S. 327, 65 USPQ 297 (1945), see MPEP 2144.07.

Claims 26 and 27 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 9 above, and further in view of Lim et al. (U.S. 2020/0310052).

Regarding Claim 26, Tai et al. in view of Chen et al. disclose the limitations of Claim 9 and further disclose the use of microbumps in the package (Tai et al., Paragraph 51) and further disclose that the system on a chip and the photonic integrated circuit are attached to the organic redistribution layer with bumps (Tai, RDL 68/64/62, bumps 66, photonic chip 50c, system on a chip 50a, Figures 1 and 44. Please note that the claim is interpreted to mean that the PIC and SOC as a group are connected to the RDL via the bumps, as it applies under broadest reasonable interpretation), but do not explicitly disclose wherein the system on a chip and the photonic integrated circuit are attached to the organic redistribution layer with microbumps. Lim et al. discloses a similar device wherein a chip and PIC are attached to a substrate with bumps (Lim et al., chip 420, PIC 406, substrate 402, bumps 422, Figures 4g and 4h). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to attach the SOC and PIC to the substrate at a fine, micro pitch using microbumps in Tai et al. in view of Chen et al., further in view of Lim et al. in order to form a device with lower cost and greater speed (Lim et al., Paragraphs 45 and 46).

Regarding Claim 27, Tai et al. in view of Chen et al., further in view of Lim et al. further disclose wherein the microbumps are secured by an underfill (Tai et al., underfill 68 and Lim et al., underfill 426, Figure 4h).

Claims 42 and 43 are rejected under 35 U.S.C. 103 as being unpatentable over Tai et al. (U.S. 2021/0157052) in view of Chen et al. (U.S. 2020/0271860) as applied to Claim 36 above, and further in view of Lim et al. (U.S. 2020/0310052).

Regarding Claim 42, Tai et al. in view of Chen et al. disclose the limitations of Claim 9 and further disclose the use of microbumps in the package (Tai et al., Paragraph 51) and further disclose that the system on a chip and the photonic integrated circuit are attached to the organic redistribution layer with bumps (Tai, RDL 68/64/62, bumps 66, photonic chip 50c, system on a chip 50a, Figures 1 and 44. Please note that the claim is interpreted to mean that the PIC and SOC as a group are connected to the RDL via the bumps, as it applies under broadest reasonable interpretation), but do not explicitly disclose wherein the system on a chip and the photonic integrated circuit are attached to the organic redistribution layer with microbumps. Lim et al. discloses a similar device wherein a chip and PIC are attached to a substrate with bumps (Lim et al., chip 420, PIC 406, substrate 402, bumps 422, Figures 4g and 4h). Therefore it would have been obvious to one of ordinary skill in the art at the time of the invention to attach the SOC and PIC to the substrate at a fine, micro pitch using microbumps in Tai et al. in view of Chen et al., further in view of Lim et al. in order to form a device with lower cost and greater speed (Lim et al., Paragraphs 45 and 46).

Regarding Claim 43, Tai et al. in view of Chen et al., further in view of Lim et al. further disclose wherein the microbumps are secured by an underfill (Tai et al., underfill 68, Lim et al., underfill 426, Figure 4h).

Response to Arguments

Applicant's arguments filed 12 February 2025 have been fully considered but they are not persuasive.

Regarding Claim 9, the Applicant argues that the Chen reference fails to disclose "an apparatus in which both the mold compound and the substrate include a cutout at the attachment of the optical fiber to the photonic integrated circuit" but does not elaborate. Applicant's arguments fail to comply with 37 CFR 1.111(b) because they amount to a general allegation that the claims define a patentable invention without specifically pointing out how the language of the claims patentably distinguishes them from the references. Therefore the argument is not persuasive. Please note that the arguments regarding the Chen reference were not made in regards to claim 36.

Applicant's arguments with respect to claims 9 and 36 regarding the "organic redistribution layer" have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Conclusion

Any inquiry concerning this communication or earlier communications from the Examiner should be directed to Abbigale Boyle whose telephone number is 571-270-7919. The Examiner can normally be reached from 11 A.M to 7 P.M., Monday through Friday.

If attempts to reach the Examiner by telephone are unsuccessful, the Examiner's supervisor, Zandra Smith, can be reached at 571-272-2429. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance form a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Abbigale Boyle

Examiner, Art Unit 2899

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Examiner, Art Unit 2899

/DALE E PAGE/
Supervisory Patent Examiner, Art Unit 2899